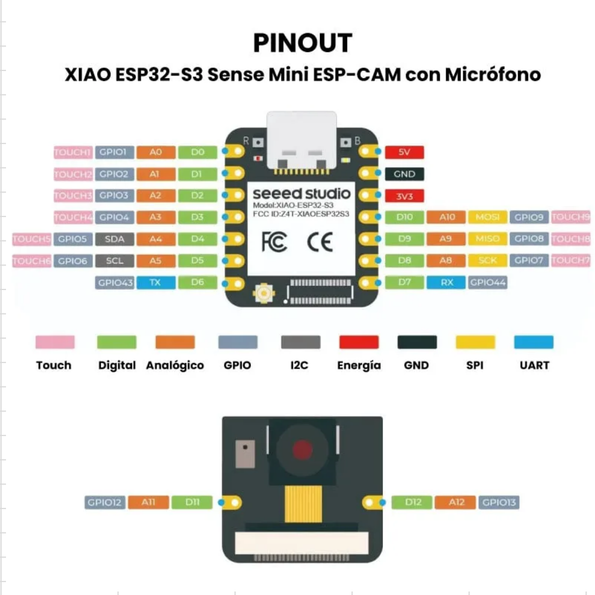
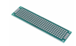
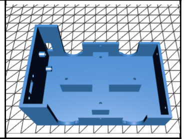
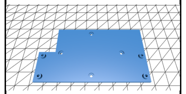
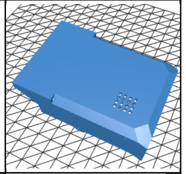
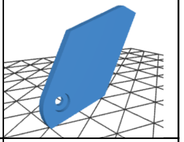
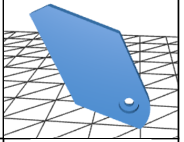
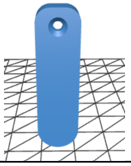
Pin No	Name	Function	PWM	Used Pins	Comment
1	D0	GPIO1 / A0	Yes	1	Digital / Analog / Touch
2	D1	GPIO2 / A1	Yes	2	Digital / Analog / Touch
3	D2	GPIO3 / A2	Yes	3	Digital / Analog / Touch
4	D3	GPIO4 / A3	Yes	4	Digital / Analog / Touch
5	D4	GPIO5 / SDA / A4	Yes	5	I2C SDA
6	D5	GPIO6 / SCL / A5	Yes	6	I2C SCL
7	D6	GPIO43 / TX	Yes	7	UART TX
8	D7	GPIO44 / RX	Yes	8	UART RX
9	5V	Power Input	No	9	5V supply
10	GND	Ground	No	10	Ground reference
11	3V3	Power Output	No	11	3.3V output
12	D10	GPIO9 / A10 / MOSI	Yes	12	SPI MOSI
13	D9	GPIO8 / A9 / MISO	Yes	13	SPI MISO
14	D8	GPIO7 / A8 / SCK	Yes	14	SPI Clock
15	D11	GPIO12 / A11	Yes	15	Camera / Digital
16	D12	GPIO13 / A12	Yes	16	Camera / Digital



No	Part No	Description	Pin No	Pin Direction	Pin Name	No of Pins	Reference	Picture
1	MG90S	Servomotor with metal gears	1	Input	PWM	3	[MG90S servomotor with metal gears - UNIT ELECTRONICS]	
			2	NO	VDD			
			3	NO	GND			
2	Mini USB Type-C Charger 5V 2A	USB-C 5V 2A charger for 1S Li-ion/LiPo with LEDs and 5V output.	1	Input	IN+	6	[Mini USB Type-C Charger 5V 2A - UNIT ELECTRONICS]	
			2	Input	IN-			
			3	Power	B+			
			4	Power	B-			
			5	Output	OUT+			
			6	Output	OUT-			
3	LiPo Battery 3.7V 1000mAh 503450	3.7V 1000mAh LiPo battery (2 pins).	1	Power	B+	2	[LiPo Battery 3.7V 1000mAh 503450 - UNIT ELECTRONICS]	
			2	Power	B-			
4	SPDT Switch	SPDT Switch (1 Pole, 2 Throw, 2.54mm pitch)	1	Input	COM	3	[INA219 Sensor Corriente 3V-5V I2C - UNIT Electronics]	
			2	Output	NO			
			3	Output	NC			
5	PCB Stripboard (2x8)	Prototype board (no fixed pinout, conductive rows)	N/A			0	[PCB Stripboard (2x8)]	
6	PH2.0 Connector	PH2.0 Connector with Cable (2-pin)	1	Power	VCC	2	[PH2.0 Connector with Cable (2-pin)]	
			2	Power	GND			
7	PH2.0 Vertical Connector	PH2.0 Vertical Connector (2-pin)	1	Power	VCC	2	[PH2.0 Vertical Connector]	
			2	Power	GND			
8	Male Header 2.54mm	Male Header 2.54mm (40 pins)	1	Signal	GENERIC	12	[Male Header 2.54mm]	

No	Description	Reference	Picture
1	Structural base of the robot that supports the servomotors, wiring system, and mechanical frame connections.	Lower Body	
2	Central mounting plate used to secure the electronic components and reinforce the overall body structure.	Plate Body	
3	Upper protective shell that encloses the ESP32S3 board, battery, and internal electronics while maintaining a compact lightweight design.	Top Body	
4	Left decorative side panel that enhances the robot's appearance and contributes to structural rigidity.	Left ear	
5	Right decorative side panel designed to complement the external chassis and improve visual symmetry.	Right ear	
6	Set of four articulated legs used for quadruped movement and servo-driven locomotion.	Limb	

Software	Purpose
ESP-IDF	ESP32 firmware development
Arduino IDE / VSCode	Embedded software development
ESP-Skainet	Voice recognition
GitHub	Version control
Fusion 360 / SolidWorks	CAD design
Cura	3D printing and slicing

Item	Quantity	Description	Comment
M2 x 6mm	6	Attach lower and upper body, power switch, and ears	Can be longer, but not shorter
M2 x 10mm	2	Charge module mounting	Can be longer, but not shorter
M2 x 4mm	8	PCB and PCB base mounting	Exact size recommended
M2 Nuts	6	Used with M2 screws	Used with M2 screwsUsed for switch, charge module, and optional extra ear support
M2.5 x 8mm	4	Servo arm screw	Default screws are too short to attach the legs